

System Test Cases: UC1 - Create and Save a Project Draft

This document defines the system test cases for **Use Case 1: Create and Save a Project Draft**. It includes the Test Case Allocation Matrix and 8 role-focused manual test cases.

Test Case Allocation Matrix (UC1)

Step	Variable / Selection	TC1 (Happy AI)	TC2 (Happy Manual)	TC3 (Blank Validation)	TC4 (Min Boundary)	TC5 (Max Boundary)	TC6 (Invalid Values)	TC7 (Upload Edge)	TC8 (Lock Verification)
B1	Action Selection	New Project	New Project	New Project	New Project	New Project	New Project	New Project	New Project
B2	Creation Method	Start with AI	Start Manually	Start with AI/Start Manually	Start Manually	Start with AI	Start Manually	Start with AI	Start Manually
B3	Project Name	"Alpha Project"	"Beta Project"	[Blank]	"A" (Min)	250 chars (Max)	"" (Only spaces)	"Uploaded Test"	"Locked Project"
B4	Expected Parts	2	5	[Blank]	1 (Min)	10 (Max)	-1 (Negative)	1	1
B5	Bin Type	First option	Last option	First option	Last option	First option	Last option	First option	First option
B6-8	Bin Dimensions (mm)	1200 x 800 x 600	800 x 600 x 400	[Blank]	100 (Min)	5000 (Max)	-10 (Negative)	1200 x 800 x 600	1200 x 800 x 600
B9	Preferred Robot	First option	Last option	First option	Last option	First option	Last option	First option	First option
B10	Additional Project Notes	"Notes text"	[Blank]	"Advice text"	"A"	1000 chars	[Special chars]	"Notes"	"Lock Check"
B11	Cell Photos	Valid PNG	Valid JPG	[No upload]	Valid PNG	Multiple PNGs	Valid JPG	Oversized / Corrupt	Valid PNG
B13	Part Name	"Part A"	"Part B"	[Blank]	"P"	250 chars	"" (Only spaces)	"Part C"	"Part D"
B14	Part Dimensions (mm)	100 * 150 * 50	200	[Blank]	1 (Min)	2000 (Max)	0 (zero)	150	150

Step	Variable / Selection	TC1 (Happy AI)	TC2 (Happy Manual)	TC3 (Blank Validation)	TC4 (Min Boundary)	TC5 (Max Boundary)	TC6 (Invalid Values)	TC7 (Upload Edge)	TC8 (Lock Verification)
B15	Part Weight (kg)	2.5	8.0	[Blank]	0.1 (Min)	100.0 (Max)	-5.0 (Negative)	2.5	2.5
B16	Part Material	"Shiny Steel"	"Cast Iron"	[Blank]	"M"	100 chars	[Numbers only]	"Aluminium"	"Steel"
B17	Desired Cycle Time (s)	15	30	[Blank]	1 (Min)	300 (Max)	0 (Zero)	15	15
B19	CAD File Available	Yes (Valid JPG)	No	Yes	No	Yes	No	Oversized (>200KB)	Yes
B21-27	Characteristics	Defaults	Custom	Defaults	Custom	Defaults	Custom	Defaults	Defaults
B32	Part Images	Valid PNG	Valid JPG	[No upload]	Valid PNG	Multiple PNGs	Valid JPG	Corrupt file	Valid PNG

Manual Test Cases

TC1: Happy Path - AI-Assisted

- **Actor:** External Partner
- **Purpose:** Verify successful project draft creation starting with the AI assistant, entering standard valid values, uploading valid PNG files, and saving.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B1 & B2: Creation Method	New Project -> Start with AI	App initializes the questionnaire with the AI Chat panel visible on the sidebar.	The app initializes two tabs i.e. AI Chat panel and AI Advice Report.	Pass	When starting with AI mode, the app initializes two tabs i.e. AI Chat panel and AI Advice report.
2	B3-B10: General Info	Project Name: "Alpha Project", Parts: 2, Bin Type: EU-Pallet, Dimensions:	Form fields accept values successfully. AI helper suggestions are responsive.	The app allows user (Integrator or Evaluator) for the verification when the user	Pass	The general project details successfully stored to the database along with the manual

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
		1200x800x600, Robot: Universal		confirms it displays "Changes applied and saved to database."		verification.
3	B11: Cell Photos	Valid PNG file	File uploads successfully and displays as a thumbnail preview.	The app asks whether to approve or not with the approve button. On 'Approve' button clicked, it displays successful message.	Pass	The user/Integrator could successfully upload their cell or cell environment condition photos in valid PNG files.
4	B13-B18: Part Details	Part Name: "Part A", Dimensions: 150, Weight: 2.5, Material: "Shiny Steel", Cycle Time: 15	Part Details section accepts values and registers shiny surface warning flag.	The app allows the user (Integrator or Evaluator) to verify provided part details first. On 'Approve button clicked', it displays the successful message.	Pass	The app successfully stored the Part1 details in the database.
5	B19-B32: Part Media	CAD: Yes (under 200KB), Part Images: Valid PNG	CAD and part images are uploaded successfully and show previews.	The app allows user (Integrator) to verify first. On 'Apply Changes' button clicked, it stores to the database successfully.	Pass	The app successfully stored the PNG CAD file for Part1 along with the verification.
6	B33: Action Selection	Save Draft	Displays success toast: "Draft saved successfully"	There is no button called 'Save Draft'. The Integrator could save it	Fail	The Integrator user has to click on dashboard tab on the

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
			ly". Redirects to Dashboard. Firestore document has status: 'draft', isLocked: false.	by clicking at the dashboard tab located in the navigation bar.		navigation bar to check their draft project.

TC2: Happy Path - Manual Draft

- **Actor:** External Partner
- **Purpose:** Verify successful manual project draft creation without AI assistance, selecting alternative (last) options, leaving optional fields blank, and uploading JPG files.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B1 & B2: Creation Method	New Project -> Start Manually	App initializes a blank questionnaire template without opening the AI Chat panel.	A blank questionnaire template is displayed to the Integrator user without the AI Chat panel along with the user contact information.	Pass	The Project & Cell Info tab UI was displayed to the Integrator user along with the user contact information where list of tabs displayed on the left side for storing other relevant details.
2	B3-B10: General Info	Project Name: "Beta Project", Parts: 5, Bin: Last option, Dimensions: 800x600x400, Robot: Last option, Notes: [Blank]	Mandatory fields accept given input; blank optional fields do not trigger validation warnings.	Mandatory fields accepted input details as expected.	Pass	The mandatory fields accepted the given input details where optional fields did not trigger any validation warnings.
3	B11: Cell	Valid JPG file	JPG image uploads	JPG images are uploaded	Pass	The app successfully

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
	Photos		successfully.	successfully.		uploaded jpg image files in the 'Upload Cell & Environment Photos' questionnaire field without displaying any warnings or unusual error message.
4	B13-B18: Part Details	Part Name: "Part B", Dimensions: 200 mm, Weight: 8.0, Material: "Cast Iron", Cycle Time (sec): 30, Notes: [Blank]	Part details are saved successfully.	All Mandatory part detail fields accept given inputs.	Pass	Mandatory fields or optional fields do not show any errors to given data.
5	B19-B32: Part Media	CAD: No, Part	CAD is set to "No". Nothing appears below this questionnaire field.	CAD questionnaire field [2.06] is set to 'No' option.	Pass	No JPG file was uploaded on this questionnaire field since nothing displayed on [2.06] field to upload, when option was set to 'No'.
6	B33: Action Selection	Save Draft	Toast: "Draft saved successfully". Project card appears on Dashboard with "Draft" status.	Same as TC1 B33: Action selection.	Fail	Same as TC1 B33: Action selection.

TC3: Blank Field & Mandatory Validation – AI Assisted/ Manual test

- **Actor:** External Partner
- **Purpose:** Verify that mandatory fields trigger validation errors when left blank and prevent saving/dashboard navigation.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B3: Project Name	[Blank]	Input box displays red warning: "Project Name is required".	(AI Assisted) In the chat box, the chat bot asks user about the Project Name. (Manual) No response	(AI Assisted/Manual) Fail	(AI Assisted) The app did not display any red warnings sign as expected. (Manual) The system did not work as expected.
2	B4: Expected Parts	[Blank]	Red warning: "Expected number of parts is required".	(AI Assisted) The chat bot asks user(Integrator) about the total number of different parts in the project. (Manual) No response	(AI Assisted/Manual) Fail	(AI Assisted) The app did not display any red warnings as expected. (Manual) The system did not work as expected.
3	B6-B8: Bin Dimensions	[Blank]	Red warning: "Bin dimensions are required".	(AI Assisted) The chat bot asks the user (Integrator) about the Bin Outer Dimensions (Length * Width * Height in mm). (Manual) No response	(AI Assisted/Manual) Fail	(AI Assisted) No red warnings appeared in the running application. (Manual) The system did not work as expected.
4	B13: Part Name	[Blank]	Red warning: "Part name is required".	(AI Assisted) No warnings displayed in the application. (Manual) No response.	(AI Assisted/Manual) Fail	(AI Assisted) Due to no similar response in the output, the output was not similar as expected. (Manual) The

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
5	B14-B15: Part Specs	[Blank]	Red warning: "Part dimensions and weight are required".	(AI Assisted) No warnings displayed in the application. (Manual) No particular response as expected.	(AI Assisted/Fail) Fail	system did not work as expected. (AI Assisted) Same as B13. (Manual) The system does not work as expected.
6	B33: Save Action	Click Save Draft	Form blocks navigation, displays validation summary, and scrolls focus on the first empty input field.	(AI Assisted/Manual) No button to save as draft.	(AI Assisted/Manual) Fail	(AI Assisted/Manual) The Integrator must click on dashboard tab on the navigation bar to check their draft project.

TC4: Minimum Allowed Boundary Values

- **Actor:** External Partner
- **Purpose:** Verify that the system accepts the minimum allowed values for all input fields.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B3: Project Name	"A" (1 character)	Value is accepted without length warnings.	Value is accepted without any warnings.	Pass	The minimum value is accepted without any warnings.
2	B4: Expected Parts	1	Value is accepted.	Value accepted.	Pass	The min value is accepted without any warnings.
3	B6-B8: Bin Dimensions	100	Minimum valid bin dimension of 100mm is accepted.	Value accepted.	Pass	The minimum value of bin dimensions of 100mm is accepted.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
4	B13: Part Name	"P" (1 character)	Part name is accepted.	Part name is accepted.	Pass	The minimum one-character Part Name is accepted.
5	B14-B15: Part Specs	Dimensions: 1mm, Weight: 0.1kg	Minimum dimensions and weight are accepted.	Minimum dimensions and weight are accepted.	Pass	The minimum dimensions and weight are accepted.
6	B17: Cycle Time	1 second	Minimum cycle time is accepted.	Minimum cycle time is accepted.	Pass	The minimum desired cycle time is accepted.

TC5: Maximum Allowed Boundary Values – AI Assisted

- **Actor:** External Partner
- **Purpose:** Verify that the system accepts the maximum allowed values and character lengths.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B3: Project Name	250 characters	Input field accepts text and shows characters remaining count.	Input field accepts text with verification to apply changes.	Pass	The max 250 characters project name is successfully stored in the database. Note: The input field did not display characters remaining count.
2	B4: Expected Parts	10	Maximum limit of parts is accepted.	Maximum limit of parts is accepted.	Pass	The maximum number of total expected parts successfully stored in the database.
3	B6-B8: Bin	5000	Maximum	Maximum	Pass	The

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
	Dimensions		dimension of 5000mm is accepted.	dimension of 5000mm is accepted after the verification.		maximum dimension of 5000mm is stored successfully in the database.
4	B10: Project Notes	1000 characters	Text field accepts up to 1000 characters without trimming.	Text field accepts up to 1000 characters without trimming.	Pass	The project notes text field of maximum 1000 characters successfully saved in the database.
5	B11: Cell Photos	Multiple PNG files	Multi-image drop zone accepts up to the maximum permitted file uploads (e.g. 5).	Multi-image (7 images uploaded) files are successfully stored in the database	Pass	The multi-images files are successfully saved in the database.
6	B13: Part Name	250 characters	Long part name is accepted.	Long 250 characters Part name is accepted.	Pass	The system successfully save the long 250 characters Part name into database.

TC6: Out-of-Bounds Input Rejections

- **Actor:** External Partner
- **Purpose:** Verify that values outside the allowed bounds are rejected with validation errors.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B3: Project Name	"" (only spaces)	Input fails validation, showing "Invalid project name".	The project name is accepted.	Fail	The Project name field did not work as expected.
2	B4: Expected Parts	0 or -3	Rejects input and displays: "Number of	The expected parts field is accepted.	Fail	The expected part did not throw any

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
			parts must be between 1 and 10".			errors on invalid inputs.
3	B6-B8: Bin Dimensions	-50 or 0	Displays validation error: "Dimensions must be positive values".	Bin dimensions are accepted.	Fail	The Bin Outer Dimensions (mm) field did not throw any error as expected on invalid value.
4	B15: Part Weight	-1.5	Displays validation error: "Weight must be greater than zero".	Part Weight is accepted.	Fail	The part weight did not throw any error as expected on invalid input data.
5	B17: Cycle Time	0 or -5	Displays validation error: "Cycle time must be greater than zero".	Cycle Time is accepted with negative values.	Fail	The cycle time field did not throw any error response on invalid input data as expected.
6	B33: Save Action	Click Save Draft	System blocks progress and refuses to commit values to Firestore.	No 'Save as draft' button to save it as draft.	Fail	Save draft action button is not there to save the project as draft. However, the Integrator User could save it as draft by clicking at Dashboard tab located at the navigation bar.

TC7: File Upload Edge Cases & Error Handling

- **Actor:** External Partner

- **Purpose:** Verify that oversized, corrupt, or invalid file uploads are blocked with descriptive error messages.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	B11: Cell Photos	Invalid file type (e.g., .pdf)	Drop zone rejects file and displays toast: "Invalid file format. Only JPG/PNG images are allowed."	The application rejected the (.pdf) file and verified with empty data.	Fail	The invalid files are got rejected during verification process without displaying any error message. The oversized file (4.86 MB) can be uploaded successfully.
2	B11: Cell Photos	Corrupt/0-byte image	System blocks upload and alerts: "File is corrupt or empty."	System blocks the upload and alerts in the console: "Fail to load resource".	Pass	The system did not allow the corrupt/0-byte image to be saved in the database.
3	B19: CAD Upload	CAD STL file > 200 KB	Drops zone blocks file, shows error message, and suggests uploading screenshots instead.	Drops zone blocks file, shows error message in the console. Store previously uploaded old PNG or JPG file to the cell photos instead.	Pass	The system does not allow Integrator to upload the STL file with size larger than 200 KB instead the system upload previously uploaded old PNG or JPG file
4	B32: Part Images	Oversized photo (> 5 MB)	Blocks file upload and alerts: "Image size exceeds the maximum	⚠️ AI error: {"error":{"code":400,"message":"The input token count exceeds the maximum	Fail	The test interrupted due to input token count exceeded the maximum number of

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
			limit of 5MB."	number of tokens allowed 1048576.", "status": "INVALID_ARGUMENT"}}		tokens allowed.

TC8: Multi-user Locking Check

- **Actors:** External Partner & Scape App Engineer
- **Purpose:** Verify that once a draft is submitted, it becomes locked and cannot be edited by the owner (External Partner) but is visible and claimable by the Scape App Engineer.

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
1	[Partner] Action	Click Submit Case	App locks the project in Firestore: <code>isLocked: true, status: 'submitted'</code> .	The 'Submit' button of the Partner UI is converted to 'Submitted (Click to Unsubmit)'. The card status in the dashboard displays the project being submitted.	Pass	The app locks the project and convert the project card as submitted in the dashboard overview.
2	[Partner] UI	Open Project Details	Form input fields are disabled (read-only). "Save Draft" and "Submit" buttons are hidden.	Form input fields are disabled (read-only) where as "Save Draft" and "Submit" buttons are hidden.	Pass	The system disabled all input fields and hides "Save Draft" and "Submit" buttons.
3	[Engineer] Action	Login & open Dashboard	Submitted project cards are visible on the Engineer's evaluator dashboard.	Submitted project cards are visible on the Engineer's evaluator dashboard.	Pass	The system displays all types of project cards on the Engineer's evaluator dashboard.
4	[Engineer] UI	Click Take Case	Engineer succeeds in	Engineer succeeds in	Pass	The engineer or evaluator

Step	Variable / Selection	Value	Expected Result	Actual Result	Pass/Fail	Comments
			self- assigning the project. takenBy field is updated in Firestore with the engineer's email.	self- assigning the project.		could successfully self-assign the project card.